What Missionary Control Left in the Language of Two 1880s isiXhosa Newspapers

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Abstract

From 1884 the missionary *Isigidimi sama-Xosa* and the African-owned *Imvo Zabantsundu* ran side by side. Did missionary control show in *what* they printed or *how* they used words? Imvo was not more political in vocabulary: of ten political concepts fixed in advance, only parliament differs (3.3×, uncorrected). Exploratory religion differs most: Isigidimi uses it four times as often, in readers' letters too, and the gap holds both in the same years and under the same editor, Jabavu. Framing evidence is weak and not specific to political words. Old spelling penalises only the African-language tokenizer. The results are consistent with control shaping what the papers printed; whether it shaped how they wrote lies below our detection floor.

Introduction and Background

Where did missionary control leave its trace in the language of the earliest isiXhosa press? In the IsiXhosa Intellectual Traditions archive (Schoots et al., 2023a), *Isigidimi sama-Xosa* was a missionary paper; *Imvo Zabantsundu*, founded in 1884 by Isigidimi's former editor J. T. Jabavu, was the first African-owned isiXhosa paper, free "from any missionary control" (Schoots et al., 2023a, p. 55). From late 1884 both had African editors, Gqoba and Jabavu (Schoots, 2023b, p. 71). Comparing the papers contrasts control, but also two editors, places and audiences, which one paper per side cannot separate. So we also hold the editor constant: Jabavu edited Isigidimi in 1881–84, then Imvo.

Communication research separates *agenda-setting*, what a paper covers (McCombs & Shaw, 1972), from *framing*, how it presents it (Entman, 1993). We measure the first as **salience**, how often a topic's words appear, and the second with word embeddings (Hamilton et al., 2016a). Such studies mostly use modern news (Card et al., 2015; Field et al., 2018); Schoots (2023b) counted root words in these papers. We add controls that make both testable, the same-editor contrast, and a letters-versus-rest split to ask *how* control worked; finally, we ask whether this history still shapes how machines read the archive (Petrov et al., 2023; Rust et al., 2021). **H1**: the papers differ in how often they use political and religious words. **H2**: Isigidimi's political words sit closer to religious contexts. **H3**: historical text and its spelling raise tokenizer cost.

Methods

Data. We use the archive's human-corrected transcriptions, not spell-checked (Schoots et al., 2023a, p. 59), from November 1884 to 1888: 286,663 tokens of Isigidimi and 1.31M of Imvo.

Salience. Concepts are word stems adapted from Schoots (2023b) (Table S1): ten political concepts fixed in advance, four neutral controls, and religion (*tixo*, *kristu*). Religion is exploratory: the archive's authors expect it (Schoots et al., 2023a, p. 57), but a classifier trained on the same years chose it (logistic regression, issue-grouped nested cross-validation, 0.83 balanced accuracy). Ratios carry issue-bootstrap CIs; Jabavu's Isigidimi runs from January 1881 to October 1884.

Threat	Response
Different years	Nov 1884–1888 only; month-matched draws
Imvo 4.6× larger	Ten Imvo samples of Isigidimi's size
Frequency effects	2000 matched random word sets (Monroe et al., 2008)
Change over time	Each paper's own drift in other years
Unstable embeddings	Tune for seed stability; average ten runs
Non-political effects	Four neutral control concepts
One paper per side	Same editor in both; place, audience unfixed

Table 1: Threats to a fair comparison, and our response.

Letters. A letter opens by addressing the editor (*nkosi mhleli … kaundifakele*, 'sir, editor … please print') and ends with a signature; 28 of 30 random detections were genuine letters (Table S5).

Framing. Embedding shift is a weak proxy for Entman's framing: it also reflects sense, grammar and topic. We train fastText embeddings (Bojanowski et al., 2017) tuned for stability across seeds, since unstable embeddings create false shifts in small corpora (Antoniak & Mimno, 2018); this favours a grammar-heavy window of 1, so we also report windows 2 and 5. A word's *profile shift* is $1 - r$ between its similarities to 500 common words in each paper (Hamilton et al., 2016b); chance comes from shuffling issues (Dubossarsky et al., 2017). The directional test, aimed at framing, asks whether political words sit nearer religious or political anchors (Table S4); to read it, we count religious and legal words near *umhlaba*.

Tokenizers. We count subwords for XLM-R (Conneau et al., 2020), Serengeti (Adebara et al., 2023) and mBERT (Devlin et al., 2019) on modern news (Adelani et al., 2023), on the same news in old spelling (per word) and on our text (per character).

Results

H1: different things, but not more politics. Of ten political concepts, only parliament differs: Imvo uses it 3.3 times as often ($p = 0.018$ against matched sets; Fig. 1); law, vote, rights, land and government do not. Exploratory religion does: Isigidimi uses it 4.2 times as often ($p = 0.020$). Neither survives correction for 15 tests ($p < 0.003$). (CIs ask if rates differ; matched sets, if a concept differs more than random words.) The religious gap holds in letters (0.33) as in the rest (0.22; Fig. 1, right). Under the same editor, Jabavu, Imvo carried religious words a third as often as Isigidimi (0.34, CI 0.24–0.47, matched-set $p = 0.038$), and parliamentary words 1.7 times as often ($p = 0.096$). Under Gqoba, Isigidimi

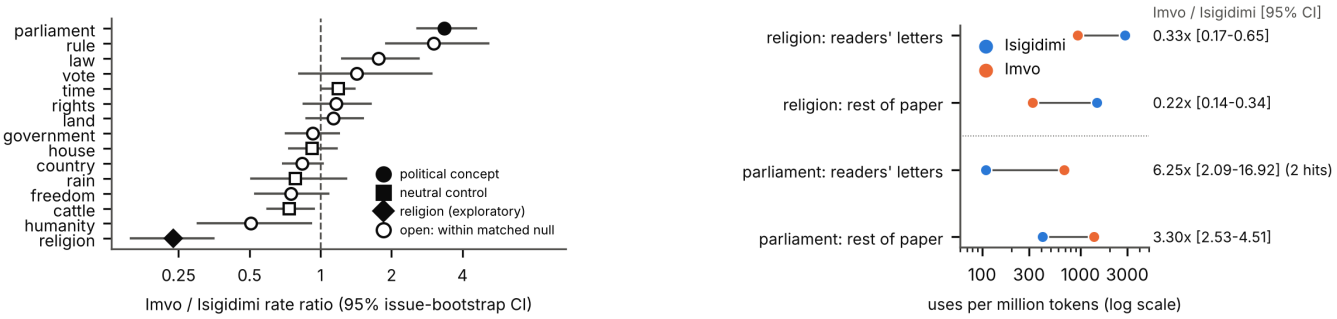


Figure 1: What the papers printed. Left: Imvo/Isigidimi rate ratio (95% CI; filled: $p < 0.05$ against matched sets, uncorrected). Right: the religious gap holds in readers' letters too.

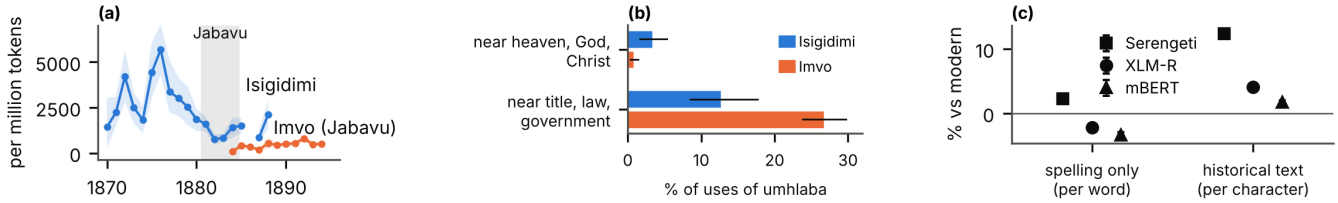


Figure 2: (a) Religious words per million tokens; grey: Jabavu edits Isigidimi, then Imvo; Gqoba edits Isigidimi from late 1884. (b) Uses of *umhlaba* near religious or legal words (95% CI). (c) Extra subwords against modern text (95% CIs smaller than markers).

carried religious words 1.45 times as often as under Jabavu ($p = 0.148$). Within Isigidimi, religious words had already fallen to 0.31 of their 1876–80 level by 1881–84, while parliament rose 50-fold and vote 12-fold ($p \leq 0.007$; Fig. 2a; Table S7).

H2: framing differences are weak and not specific. Word use differs overall (profile shift 0.348 vs 0.230 by chance); no concept stands out: parliament reaches the 96th percentile of matched sets, the control *cattle* 90th–99th (Fig. S1). Political words lean towards religious anchors in Isigidimi in all five settings (Δ +0.44 to +0.76 z-units; Bonferroni-significant in three; Table S2). But without *land* the lean is rarely significant ($p = 0.010$–0.256); the control *house* leans further (+0.84 to +1.51); dropping the anchor *endlu* ('in the house') changes neither; and *umhlaba* sits near religious words in 3.4% of Isigidimi's uses but 0.9% of Imvo's (Fig. 2b). A one-word concept must shift by 0.15 (1.645 matched-set SDs) to be detected, more than the overall gap (0.118).

H3: historical text costs more. Per character, our text needs 12.4%, 4.1% and 1.7% more subwords with Serengeti, XLM-R and mBERT; old spelling alone raises only Serengeti's cost (2.2% per word; Fig. 2c).

Discussion

Is it control, or time? (H1). Religious words were falling in Isigidimi anyway, so we triangulate over a 2×2 of control and editor, one cell missing. Imvo vs Gqoba's Isigidimi fixes the years (0.24); Jabavu's two papers fix the editor (0.34). Both show the gap: two contrasts with different confounds agree in direction (their p-values are uncorrected). Jabavu's Isigidimi also predates the years in which the classifier chose religion, so this contrast partly replicates it on held-out text. Were the gap only the trend, Isigidimi would have kept falling after 1884; it did not (1.45). A Jabavu-era Imvo as religious as his Isigidimi would have refuted control, and a further fall under Gqoba would have undermined it; neither occurred. Politics instead followed the times, rising within Isigidimi before Imvo existed. Whether Jabavu was right to leave is beyond our data; leaving brought less religion, not clearly more politics. The gap reached beyond editors and is not only a matter of which sections each paper ran: Isigidimi's letter-writers used religious words three times as often as Imvo's, so the papers drew on, or chose, different writers. One wrote that *ilizwi alizanga ukuba silipate, lize ukuba lisipate* ('the Word did not come for us to carry it; it came to carry us', or 'rule us'; 1 Nov. 1884; our gloss, for speaker review).

What the embeddings see (H2). The framing signal looks like a difference in meaning, not slant: Isigidimi more often uses *umhlaba* for the earth, Imvo for land under law, which alone would move the word in each model. The detection floor makes the null a bound: large framing differences are ruled out, small ones are not.

Responsible NLP (H3). History also reaches today's tools: written isiXhosa took shape in the mission press (cf. Gilmour, 2007), and its old spelling penalises only the African-language model, perhaps because Serengeti's vocabulary has twice XLM-R's share of *th*/*kh*/*ph* pieces (1.9% vs 0.9%); a one-model hypothesis. A normalised working copy must not replace the source, as the mapping is lossy (old *t* may be *t* or *th*). Our results describe two papers, not isiXhosa speakers; cultural readings without speaker review would be extractive (Bird, 2020).

Limitations and conclusion. Two papers, one window, a missing year (1886), an incomplete 2×2, a heuristic letter split, a domain mismatch, unreviewed glosses (Table S1). The evidence fits control shaping what the papers printed, down to their letter-writers; any effect on political language is below our detection floor.

References

Adebara, I., Elmadany, A., Abdul-Mageed, M., & Alcoba Inciarte, A. (2023). SERENGETI: Massively multilingual language models for Africa. In *Findings of the Association for Computational Linguistics: ACL 2023.*

Adelani, D. I., et al. (2023). MasakhaNEWS: News topic classification for African languages. In *Proceedings of IJCNLP-AACL 2023*.

Antoniak, M., & Mimno, D. (2018). Evaluating the stability of embedding-based word similarities. *Transactions of the Association for Computational Linguistics*, *6*, 107–119.

Bird, S. (2020). Decolonising speech and language technology. In *Proceedings of COLING.*

Bojanowski, P., Grave, E., Joulin, A., & Mikolov, T. (2017). Enriching word vectors with subword information. *Transactions of the Association for Computational Linguistics*, *5*, 135–146.

Card, D., Boydstun, A. E., Gross, J. H., Resnik, P., & Smith, N. A. (2015). The media frames corpus: Annotations of frames across issues. In *Proceedings of ACL-IJCNLP.*

Conneau, A., Khandelwal, K., Goyal, N., et al. (2020). Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics.*

Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT* (pp. 4171–4186).

Dubossarsky, H., Weinshall, D., & Grossman, E. (2017). Outta control: Laws of semantic change and inherent biases in word representation models. In *Proceedings of EMNLP.*

Entman, R. M. (1993). Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, *43*(4), 51–58.

Field, A., Kliger, D., Wintner, S., Pan, J., Jurafsky, D., & Tsvetkov, Y. (2018). Framing and agenda-setting in Russian news: A computational analysis of intricate political strategies. In *Proceedings of EMNLP.*

Gilmour, R. (2007). Missionaries, colonialism and language in nineteenth-century South Africa. *History Compass*, *5*(6), 1761–1777.

Gonen, H., Jawahar, G., Seddah, D., & Goldberg, Y. (2020). Simple, interpretable and stable method for detecting words with usage change across corpora. In *Proceedings of ACL*.

Hamilton, W. L., Leskovec, J., & Jurafsky, D. (2016a). Diachronic word embeddings reveal statistical laws of semantic change. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics* (pp. 1489–1501).

Hamilton, W. L., Leskovec, J., & Jurafsky, D. (2016b). Cultural shift or linguistic drift? Comparing two computational measures of semantic change. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 2116–2121).

McCombs, M. E., & Shaw, D. L. (1972). The agenda-setting function of mass media. *Public Opinion Quarterly*, *36*(2), 176–187.

Monroe, B. L., Colaresi, M. P., & Quinn, K. M. (2008). Fightin' words: Lexical feature selection and evaluation for identifying the content of political conflict. *Political Analysis*, *16*(4), 372–403.

Petrov, A., La Malfa, E., Torr, P. H. S., & Bibi, A. (2023). Language model tokenizers introduce unfairness between languages. In *Advances in Neural Information Processing Systems.*

Rust, P., Pfeiffer, J., Vulić, I., Ruder, S., & Gurevych, I. (2021). How good is your tokenizer? On the monolingual performance of multilingual language models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics*.

Schoots, J. (2023b). Analyzing political formation through historical isiXhosa text analysis: Using frequency analysis to examine emerging African nationalism in South Africa. In *Proceedings of the Fourth Workshop on Resources for African Indigenous Languages (RAIL 2023)* (pp. 65–75).

Schoots, J., Ngwendu, A., De Wet, J., & Muftic, S. (2023a). IsiXhosa intellectual traditions digital archive: Digitizing isiXhosa texts from 1870-1914. In *Proceedings of the Fourth Workshop on Resources for African Indigenous Languages (RAIL 2023)* (pp. 54–64).

Statement on the use of generative AI (Faculty of Science AI Ethics Statement)

Generative AI (Claude, by Anthropic, used through the Claude Code tool) helped to write the analysis code (data preparation, embedding experiments, statistics, figure scripts), to draft and edit the text and LaTeX source, and to fill in the checklist below from the course template. Generative AI was also used to review drafts and suggest revisions; suggested changes were checked against the code outputs and the cited sources before being kept. All numbers in the paper are produced by the submitted code and were read from its output files. The authors take full responsibility for all code and text submitted, including the accuracy of every claim and citation.

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Supplementary material

S1. Concepts, stems and the word forms they match (1 Nov 1884–1888, both papers)

Stems are matched as substrings of surface forms; following Schoots (2023b) we inspected forms for false positives (our first *law* stem also matched *inteto*, speech, and was dropped) but did not hand-filter them, so run-together forms (e.g. *rulumenteningomsebenzi*) and missed variants remain. All stems, forms and glosses await review by a first-language isiXhosa speaker; e.g. *rulumente* is glossed 'government' here but 'parliament' by Schoots et al. (2023a, p. 62). Political concepts were fixed before testing; religion was suggested by the classifier and is exploratory.

Concept	Type	Stems	Most frequent matched forms (count; number of distinct forms)
parliament	political	palamente	palamente (1754), ipalamente (47), palamenteni (22), epalamente (14), bapalamente (1), nepalamente (1), yepalamente (1); 11 forms
vote	political	voti	ivoti (301), voti (217), evotini (136), abavoti (98), ngevoti (55), levoti (22), nevoti (22); 115 forms
government	political	rulumente	rulumente (1882), rulumenteni (7), urulumente (5), serulumente (1), rulumenteningomsebenzi (1), zikarulumente (1), rulumenteayiko (1); 8 forms
rule	political	mbuso	zombuso (169), umbuso (76), mbuso (60), yombuso (42), kombuso (37), lombuso (28), kulombuso (23); 43 forms
law	political	mteto	umteto (724), lomteto (475), mteto (406), ngomteto (167), ngumteto (94), komteto (76), nomteto (76); 82 forms
rights	political	lungelo	amalungelo (315), ilungelo (288), lungelo (113), nelungelo (54), namalungelo (50), lilungelo (39), elilungelo (37); 115 forms
country	political	lizwe	lizwe (646), ilizwe (292), kwelilizwe (201), elilizwe (101), elizwe (96), belilizwe (76), yelilizwe (52); 78 forms
land	political	mhlaba	umhlaba (836), mhlaba (208), lomhlaba (156), nomhlaba (105), komhlaba (99), somhlaba (95), yomhlaba (94); 135 forms
freedom	political	kululek	inkululeko (27), kukululekile (24), ukululekile (13), ngenkululeko (10), abakululekileyo (8), nkululeko (8), nenkululeko (7); 92 forms
humanity	political	buntu	ubuntu (45), yobuntu (14), nobuntu (8), sobuntu (8), buntu (7), lobuntu (7), zobuntu (6); 29 forms
time	neutral	xesha	ixesha (787), ngexesha (364), xesha (298), amaxesha (243), lixesha (191), ngamaxesha (149), zamaxesha (114); 207 forms
house	neutral	ndlu	indlu (719), izindlu (328), kwindlu (276), ndlu (234), yendlu (227), ngendlu (133), nendlu (130); 193 forms
rain	neutral	mvula	imvula (267), mvula (31), yemvula (23), ezimvula (17), lemvula (16), yimvula (14), zimvula (10); 45 forms
cattle	neutral	nkomo	inkomo (555), zenkomo (258), nkomo (193), yenkomo (93), nenkomo (75), ngenkomo (35), lenkomo (33); 79 forms
religion	exploratory	tixo, kristu	tixo (684), kristu (207), makristu (7), kristukazi (7), mkristu (3), amakristu (2), lobukristu (1); 16 forms

S2. Secondary framing tests across settings

Percentile of the observed shift among 2000 frequency-matched word sets, and one-sided p (uncorrected). Jaccard = shift in a word's ten nearest neighbours (neighbour overlap; Gonen et al., 2020). P+V = *palamente* + *voti*; P+V+R adds *rulumente* (uncertain meaning). Directional Δ: Isigidimi minus Imvo, religious minus political similarity (in z-score units), political concepts, two-sided p against matched sets; the last columns give the control *house*. Three settings survive Bonferroni over five ($p < 0.01$): window 1, window 2, and window 2 with forms joined. Joining partly disjunctive forms (*uku ba*) leaves the global profile gap unchanged. Matched sets are re-drawn for each test; without *land* the pooled set is smaller, so lower power may contribute to the higher p.

Setting	P+V profile	P+V Jaccard	P+V+R profile	P+V+R Jaccard	Δ	Δ p	p w/o land	house Δ	house p
window 1 (tuned)	92 (p=0.083)	96 (p=0.039)	92 (p=0.077)	99 (p=0.007)	+0.68	<0.001	0.010	+1.08	<0.001
window 1, forms joined	99 (p=0.010)	95 (p=0.053)	100 (p=0.001)	100 (p=0.002)	+0.44	0.016	0.088	+1.15	<0.001
window 2	86 (p=0.139)	94 (p=0.061)	90 (p=0.096)	99 (p=0.010)	+0.76	<0.001	0.045	+1.34	<0.001
window 2, forms joined	91 (p=0.094)	98 (p=0.023)	99 (p=0.011)	100 (p=0.002)	+0.58	0.009	0.064	+1.51	<0.001
window 5	62 (p=0.382)	98 (p=0.016)	81 (p=0.189)	100 (p=0.002)	+0.66	0.015	0.256	+0.84	0.061

S3. Era drift: does a concept's rate move across eras within one paper?

Per-million-token rates and ratios (later/earlier, or Imvo/Isigidimi) with issue-bootstrap 95% CIs.

Concept	Comparison	First	Second	Ratio (95% CI)
parliament	Isigidimi: later / earlier era	214	394	1.84 (0.96–4.98)
parliament	Imvo: later / earlier era	1320	758	0.57 (0.50–0.66)
parliament	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	214	758	3.54 (1.98–9.58)
government	Isigidimi: later / earlier era	1667	1088	0.65 (0.50–0.85)
government	Imvo: later / earlier era	1210	452	0.37 (0.31–0.45)
government	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	1667	452	0.27 (0.22–0.34)
vote	Isigidimi: later / earlier era	72	614	8.55 (3.51–18.94)
vote	Imvo: later / earlier era	799	921	1.15 (0.88–1.51)
vote	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	72	921	12.82 (7.85–25.92)
religion	Isigidimi: later / earlier era	2466	1556	0.63 (0.46–0.84)
religion	Imvo: later / earlier era	362	547	1.51 (1.15–2.03)
religion	Imvo 1889-94 / Isigidimi 1876-83 (held-out)	2466	547	0.22 (0.18–0.28)

S4. Anchors for the directional test

Anchors are the 15 top-PMI neighbours of each seed pair in held-out text (Isigidimi 1876-83 + Imvo 1889-94), frozen before testing. Religious (15 words; seeds *tixo*, *kristu*): *sikelela*, *sindisa*, *ukumkanikazi*, *inkosazana*, *buka*, *oyingcwele*, *lilizwi*, *msindisi*, *nelizwi*, *kwelizwi*, *wanga*, *lika*, *yesu*, *izono*, *yehova*. Political (15 words; seeds *palamente*, *voti*): *emqulwini*, *wonyulo*, *lonyulo*, *elugqatsweni*, *eziqingata*, *lokunyula*, *kwesase*, *lokuvota*, *endlu*, *kwamagama*, *ukumela*, *izayo*, *ngalomteto*, *nawupina*, *kwabase*. Seed words cannot themselves be tested (Fig. S1). Political anchors include *endlu* ('in the house'), which might have favoured the neutral control *house*; re-running without *endlu* leaves both the political lean and *house*'s lean in place (outputs/framing_matched_tests_noendlu.csv), and *ngalomteto*, which contains the *law* stem. Glosses are unverified: two religious anchors (*ukumkanikazi*, *inkosazana*) may reflect royal rather than religious register, and several anchors on both sides (*lika*, *buka*; *nawupina*, *kwabase*, *izayo*) may be grammatical rather than topical, consistent with narrow-window PMI capturing grammar. All await speaker review.

S5. Readers' letters versus the rest of each paper

Letters are detected by src/12_letters.py: a letter opens by addressing the editor (vocative *mhleli* or *mcokeli*, not *u/ku/ngu mhleli*, which refer to him) or by asking for insertion (*ndifakele*, *kaundifakele*), and ends at the signature (a run of very short lines). A hand check of 30 random detections found 28 genuine letters; the errors were usifake umkala (horse advert: 'put a bit on it'); no mhleli wepepa lama bhulu ('and the editor of the Boer paper'). Recall was not measured. Imvo is restricted to the months in which Isigidimi has text; CIs from 4000 issue resamples; rates are not tested against matched sets. Hits = concept uses.

Part	Concept	Isig. tokens	Isig. hits	Imvo tokens	Imvo hits	Imvo/Isig.	95% CI
letters	parliament	18,344	2	70,475	48	6.25	2.09–16.92
letters	religion	18,344	52	70,475	66	0.33	0.17–0.65
letters	vote	18,344	4	70,475	58	3.77	1.04–18.93
letters	government	18,344	13	70,475	115	2.30	1.14–7.18
rest of paper	parliament	268,319	111	936,998	1280	3.30	2.53–4.51
rest of paper	religion	268,319	394	936,998	306	0.22	0.14–0.34
rest of paper	vote	268,319	172	936,998	821	1.37	0.79–2.83
rest of paper	government	268,319	299	936,998	902	0.86	0.67–1.13

S6. Which sense of *umhlaba*?

For every use of the land stem (*mhlaba*) we look at the 10 tokens either side. Religious cues: *izulu*, *lizulu*, *nezulu*, and words containing *zulwini*, *tixo*, *kristu*, *yesu*, *msindisi*, *ngcwele* (bare *zulu* is excluded: *kwa zulu* is Zululand). Property/political cues: words containing *taitile*, *mteto*, *rulumente*, *palamente*, *mantyi*, *fama*, *voti*, *nyulo*, *rafu*, *lokishi*. Most uses have neither cue. Example lines for a speaker to check are in outputs/land_kwic.md.

Paper	Cue	Uses	Share	95% CI
Imvo	religious	1408	0.9%	0.4–1.4%
Imvo	property/political	1408	26.8%	23.8–29.7%
Isigidimi	religious	353	3.4%	1.7–5.3%
Isigidimi	property/political	353	12.7%	8.5–17.7%

S7. Same editor, different control; same control, different editor

Cells (editors after Schoots, 2023b, p. 71): Jabavu edits Isigidimi (Jan 1881–Oct 1884); Jabavu edits Imvo (Nov 1884–1888); Gqoba edits Isigidimi (Nov 1884–1888); earlier editors of Isigidimi (1876–80). Ratios of rates per million tokens; 95% CIs from 4000 issue resamples; p: two-sided, against 2000 matched sets of random words (src/15_editor_control.py). Every contrast is also a contrast in time; *rule* has only 9 uses in Jabavu's Isigidimi, so its large ratios are not interpreted.

Concept	Jabavu: Imvo / Isigidimi			Isigidimi: Gqoba / Jabavu			Isigidimi: 1881–84 / 1876–80		
	ratio	95% CI	p	ratio	95% CI	p	ratio	95% CI	p
parliament	1.69	1.23–2.54	0.096	0.50	0.34–0.80	0.041	49.86	14.59–403.58	<0.001
vote	3.07	1.98–5.10	0.012	2.36	1.04–4.55	0.025	11.63	4.42–104.17	0.007
government	0.85	0.65–1.13	0.525	0.77	0.56–1.04	0.260	0.88	0.65–1.18	0.599
rule	19.47	9.52–76.87	0.002	8.28	3.61–31.65	0.015	0.38	0.09–1.36	0.227
law	1.36	1.07–1.71	0.210	0.80	0.54–1.17	0.378	1.31	0.97–1.77	0.343
rights	0.93	0.76–1.16	0.801	0.82	0.59–1.12	0.482	2.91	2.09–4.10	0.006
country	1.08	0.92–1.28	0.780	1.13	0.91–1.42	0.620	0.74	0.59–0.92	0.249
land	1.10	0.91–1.34	0.710	0.95	0.70–1.29	0.834	0.94	0.76–1.16	0.801
freedom	0.68	0.51–0.91	0.444	0.95	0.64–1.39	0.903	1.52	1.01–2.37	0.402
humanity	0.97	0.58–1.88	0.961	1.88	0.97–3.87	0.213	0.97	0.44–2.03	0.966
religion	0.34	0.24–0.47	0.038	1.45	1.06–1.98	0.148	0.31	0.23–0.42	<0.001

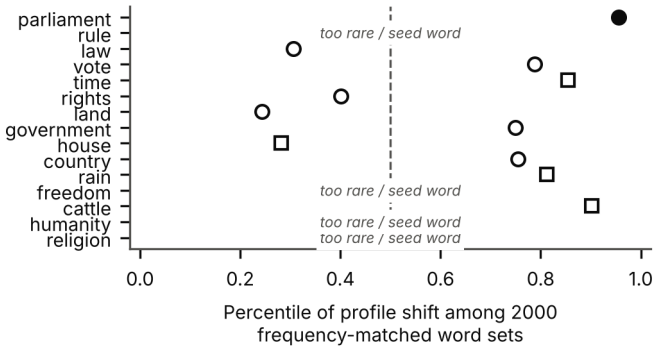


Figure S1: Profile-shift percentile of each concept among matched sets (markers as in Fig. 1, left); 'too rare' = too few occurrences, 'seed word' = anchor source, untestable.

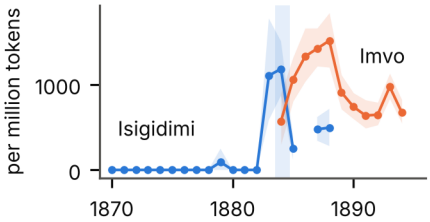


Figure S2: Parliamentary vocabulary per million tokens; it enters Isigidimi in 1883 (light blue: twice-monthly issues, Schoots et al., 2023a, p. 59).